

The Double Ladder Theorem

The discriminant group of the Néron–Severi lattice of a prime-degree Fermat surface — structure, not just order

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5 June 2026 · version 4, 30 September 2026 (see the notes below) · Companion to The Watermark Theorem (same repository). Computations: MacBook Air M2, 8 GB RAM, single thread, 25% CPU.

Version 3 (30 September 2026). The statement is unchanged. Corrections, after reading the sources in the original: (1) Aljovin–Movasati–Villafior [6, Table 1] list the elementary divisors not only at $m = 5, 7$ but also at $m = 11$ and $m = 13$, and all four rows agree with the theorem; so the $m = 11$ value, although predicted before it was computed, was already in print. (2) Shioda [7, Questions 7.2 and 7.4] asked for the determinant of $NS(S_m)$, not for its discriminant group; the Watermark Theorem answers his question for prime degree, and the present theorem goes further. (3) The 170 GB of swap in [6] were for their fourfold cell $(n, d) = (4, 6)$. **Status of the proof.** §4 and §6 below contain four steps confirmed at three or four primes and not written for general m (see the Remark in §4). A different proof, written for every prime — the order from the Watermark, the exponent m^2 from an explicit integer identity $G Y G = 6m^2 G$, and $\text{rank}(G \bmod m) = 12(m - 3)$ for $m \geq 7$ (26 at $m = 5$) — has been checked completely in Lean 4 with Mathlib on 30 September 2026, together with the Watermark. Its Lean certificate is being prepared for this repository.

Version 4 (30 September 2026, later the same day). The statement is unchanged.

1. Which proof is complete. The Abstract and §1 present the proof of this paper as complete. Four of its steps, in §4 and §6, are confirmed at three or four primes and not written for general m (the Remark in §4; the note to version 3). The theorem is proved for every prime by a different proof: the order from the Watermark Theorem, the exponent m^2 from the integer identity $G Y G = 6m^2 G$, and $\text{rank}(G \bmod m) = 12(m - 3)$ for $m \geq 7$ (26 at $m = 5$). That proof is written out in *The Double Ladder Theorem — a pencil proof* (version 2), and it is the one certified in Lean.

2. Verified in Lean. The Lean 4 certificate, `LEAN_CERTIFICATE_DOUBLE_LADDER_v2`, announced in the note to version 3, is published. For every prime $m \geq 7$, Lean proves $V^*/V \cong (Z/m)^{3m^2-24m+59} \times (Z/m^2)^{3m-16}$, and it proves the case $m = 5$ separately. The proofs use only the three standard axioms. An independent cold reader found no fatal error and no gap. That the Lean matrix is the intersection matrix of the lines is proved by hand and checked by computer, not formalized. The certificate, the pencil proof, the Lean project and all the evidence are in the repository github.com/tretoef-estrella/watermark-theorem.

Abstract

For the Fermat surface $S_m \subset P^3$ of prime degree $m \geq 5$, we determine the discriminant group of the Néron–Severi lattice $V = NS(S_m)$ completely:

Theorem (Double Ladder). $V^*/V \cong (Z/m)^{3m^2-24m+59} \times (Z/m^2)^{3m-16}$ for every prime $m \geq 7$, and $V^*/V \cong (Z/5)^{10} \times (Z/25)$ for $m = 5$.

The Watermark Theorem (companion document) gives the *order* of this group, $m^{3(m-3)^2}$; the present theorem gives its *structure* — the Watermark answers, for prime degree, the determinant question of Shioda (1987, Question 7.4), and the two together give the complete elementary-divisor profile of the lattice. Since $H^2(S_m, Z)$ is unimodular, the transcendental lattice T_m carries the same discriminant group. The proof is a chain of explicit lattice computations on the integer Gram matrix of the $3m^2$ lines, organised by two new structural objects: **nine affine "shadow" classes** whose evaluation matrix has determinant **6** — a unit for every prime $m \geq 5$ precisely because $\gcd(m, 6) = 1$, the same hypothesis under which the lines generate NS — and a **rank ladder** $\text{rank}(Q|m^K) = 12(m-3) - 3K(K+1)$ over a moment filtration, whose rung $K = 2$ closes the count. Every law was measured at three primes before being proven, the $m = 11$ profile was predicted in writing before it was computed (it is also a row of [6, Table 1]), and every lemma was reproduced independently in a second environment.

1. The question and the answer

Shioda, closing his 1987 paper on Jacobi sums [7], asked (Questions 7.2 and 7.4) for the determinant of the Néron–Severi lattice of the Fermat surface of prime degree: is it $m^{3(m-3)^2}$? For $\gcd(m, 6) = 1$ the lattice is generated by the $3m^2$ lines on the surface (Schütt–Shioda–van Luijk [4], completed by Degtyarev [5]), so the question is in principle a finite computation for each m — and Aljovin–Movasati–Villaflor [6] carried that computation out cell by cell within their hardware’s reach, tabulating the elementary divisors for the surfaces of degree up to 14 (their Table 1 contains all three anchors below, and also $m = 13$), with swap memory raised to 170 GB for their fourfold cell (4, 6), while stating that their largest cells (the linear-cycle lattices behind their Table 2) were beyond computation. What no source provides — neither there nor anywhere we could locate — is a *closed law* valid for every prime, or a *proof*: each published value is an isolated machine verdict. The present theorem supplies both, for every prime, beyond every table (nothing beyond $m = 14$ exists in print).

The Watermark Theorem proved the order: $|V^*/V| = m^{3(m-3)^2}$ for every prime $m \geq 5$. Since the exponent of the group divides m^2 (§3 below), the group is $(Z/m)^a \times (Z/m^2)^b$ with $a + 2b = 3(m-3)^2$, and the entire remaining question is **one integer**:

$$b = \#(\text{cyclic factors of order } m^2).$$

The Double Ladder Theorem asserts $b = 3m - 16$ for $m \geq 7$ and $b = 1$ for $m = 5$ — whence $a = 3m^2 - 24m + 59$ (resp. 10).

Three anchors, all rows of [6, Table 1]. (1) At $m = 5$ the profile $(Z/5)^{10} \times Z/25$. (2) At $m = 7$: $(Z/7)^{38} \times (Z/49)^5$. (3) At $m = 11$: $(Z/11)^{158} \times (Z/121)^{17}$, written down as a prediction *before* our computation and met exactly on all counts. (The row $m = 13$ of [6], $(Z/13)^{254} \times (Z/169)^{23}$, agrees as well.)

2. Notation and substrate (results carried in from the Watermark chain)

$A = Z^{3m^2}$ is the free module on the lines, G its integer Gram matrix, $\varphi: A \twoheadrightarrow V$ the class map with kernel K (saturated). The character orbits split the configuration into $3(m-2)$ **pieces**: for each orbit O with pencil L , $T(O) \subseteq S(O) := V \cap (\text{piece})$ are the trace lattice and its saturation. From the Watermark campaign and the first phase of this operation, the following are **theorems** (proofs in the companion document and in the assembly; instruments GOLLUM_PROBE_1–3):

- **Flat law.** $\text{disc } S(O) = m^{2m-5}$ for every piece, every prime $m \geq 5$; the dual group $S(O)^*/S(O)$ is the cyclic module O/λ^{2m-5} of the cyclotomic ring ($\lambda = (1-\zeta)$), with abelian profile $(Z/m)^3 \times (Z/m^2)^{m-4}$ — by the λ -arithmetic $2m-5 = (m-1) + (m-4)$. (Cyclicity in one line: $S(O)^*/S(O)$ is a subquotient of $T(O)^*/T(O)$, which is cyclic over O by the circulant presentation of the trace lattice — Lemma T of the Watermark chain — and a subquotient of a cyclic module is cyclic.)
- **Glue law.** $H := V/L$ for $L := \oplus_0 S(O) \oplus Ze$ is elementary m -torsion of dimension $g = (3m^2-9m+4)/2$ over F_m .
- **The cap.** $[S(O) : T(O)] = m^m$ exactly — the saturation steps are the **jewel identity** $t_r + N_j = m \cdot \text{pb}_L(e_r)$ (so the saturated generators are fiber-indicator classes, computable in A with no kernel extraction) and the linear weight $\Sigma_r r \cdot v_r \in m^2V$, with the cap proven by a five-pencil polynomial analysis in the register of the Watermark's Lemma C.
- **Lemma D (dictionary).** K saturated $\Rightarrow A \cong K \oplus V \Rightarrow \text{coker}(G) \cong (V^*/V) \oplus Z^{9m-7}$: the Smith normal form of the explicit Gram matrix computes the discriminant group of NS exactly. (This is what makes the sealed $m = 11$ measurement *theorem-backed*.)

Write $D := L^*/L$ (exponent m^2), $t := \#(m^2\text{-cyclics of } D) = 3(m-2)(m-4)$, $H_{\text{top}} := H \cap mD$, $\alpha := \dim_{\{F_m\}} H_{\text{top}}$, and let B_2 be the perfect F_m -bilinear form on $mD \cong F_m^t$ induced by the m^2 -part of the discriminant pairing.

3. Lemma E — the height formula

Lemma E. $b = t - 2\alpha + \rho_2$, where $\rho_2 := \text{rank}(B_2|_{H_{\text{top}}})$.

Proof sketch (full text in the assembly). By overlattice theory $V^*/V \cong H^\perp/H$ inside D , and $b = \dim_{\{F_m\}} m \cdot (H^\perp/H) = \dim mH^\perp - \dim(mH^\perp \cap H)$. The key computation is the radical of the $(1/m)$ -pairing on $D[m]$: it equals mD , because the induced form on $D[m]/mD$ — per piece, the three top λ -grades of the cyclic module O/λ^{2m-5} , plus the line $F_m \cdot e$ — is the standard duality of a cyclic module: grade i pairs with grade $2-i$, an anti-triangular matrix with unit antidiagonal, and e contributes the unit $(1/m) \cdot m$. Hence $z \in mH^\perp \iff B_2(z, H_{\text{top}}) = 0$, giving $\dim mH^\perp = t - \alpha$ by perfectness of B_2 ; and $mH^\perp \cap H$ is exactly the B_2 -radical of H_{top} , of dimension $\alpha - \rho_2$. Subtract. ■

Gate of record. An end-to-end instrument computing $H^\perp/H$ directly — knowing nothing of the formula — returns $b = 1$ ($m = 5$) and $b = 5$ ($m = 7$), the values the formula yields (GOLLUM_PROBE_7).

The theorem is now two invariants: α and ρ_2 .

4. First invariant — the nine affine shadows and the autograph determinant 6

The affine shadow law. The class of the line at position (k, l) of family j , taken modulo mD , is $u_j + k \cdot v_j + l \cdot w_j$ — an *affine* function of position. The nine classes $\{u_j, v_j, w_j\}$ are linearly independent for every prime $m \geq 5$; hence the glue's image in D/mD is exactly their 9-dimensional span, and $\alpha = g - 9 = (3m^2 - 9m - 14)/2$ for all $m \geq 5$.

Affineness follows from the fiber identities of the Watermark's Lemma R: a line's pairing vector against a piece basis is (cross-family constants) + $(m \cdot \delta$ terms); the deep $m \cdot \delta$ parts vanish modulo mD , and what remains — constants and values of the linear function y_{lin} — is affine in (k, l) . Affine maps in two variables over three families cap the span at 9. Independence is sealed by an explicit dual evaluation: because the piece Gram is $\equiv 0 \pmod{m}$ (entries $2m - m^2\delta$), the classes $pb_L(e_r)/m$ and a_0/m are canonical order- m functionals; three "e-readers" (which annihilate all six increments) and six " σ -readers" (which read the pencil slope (a, b) on the increments) assemble a block-triangular 9×9 evaluation matrix with

determinant = 6, the same integer at every prime (measured $1 \equiv 6 \pmod{5}$, then 6, 6, 6 at $m = 5, 7, 11, 13$) —

a unit for every prime $m \geq 5$ precisely because $\gcd(6, m) = 1$: the seal closes on the same hypothesis ($\gcd(m, 6) = 1$) under which the lines generate NS. Verified entrywise for every line at two primes; the determinant at four.

Remark (status of the uniformity, stated for the record). The block-triangular reduction and the annihilation of the increments by the e-readers are proven from the displayed identities; the determinant is therefore a fixed integer provided the reader-evaluation entries are independent of m , as the displayed values (the family-pair constants $2/1/0$ and the pencil slopes) indicate. The fully written entry table — the one step separating "fixed integer, confirmed at four primes" from a self-contained symbolic evaluation — is on the campaign's verification queue and is the standard this document holds itself to; the four-prime confirmation ($m = 5, 7, 11, 13$, the integer 6 each time) is reported as exactly what it is.

5. The form is the configuration — B_2 is the line Gram

By §4, H_{top} is described explicitly: combinations whose coefficient function on each family has **zero mass**, **zero k-moment**, **zero l-moment** modulo m . For such a combination c , the fiber identities force $\langle v_c, L \rangle \in m\mathbb{Z}$, so $v_c/m \in L^*$ is a legitimate representative of $(1/m)$ -classes — and the abstract discriminant form collapses:

$B_2(z, z') = c^T G c' \pmod{m}$ — the line-configuration Gram matrix itself.

No dual frames, no normalizations: the form the theorem needs is the matrix the configuration *is*. Measured rank: $8 / 30 / 78$ at $m = 5 / 7 / 11$, each value declared before the run (GOLLUM_PROBE_8/10).

6. Second invariant — the rank ladder

Modulo m , G is pure affine-line incidence: the diagonal blocks are exactly $1(k=k') + 1(l=l')$ — in one line from the Watermark's Lemma P: same-family pairings are $-(m-2) \equiv 2$ (self), 1 (one position shared), 0 (disjoint), which is $1(k=k') + 1(l=l') \bmod m$ — and the cross blocks are the single affine lines $\{k-l\}$, $\{k+l\}$, and one twisted line. Consequently the quadratic form $Q(c) = c^T G c \bmod m$:

1. **factors exactly through twelve one-variable "shadows"** — the pushforwards of each family's coefficient function along the four pencil directions $\{k, l, k-l, k+l\}$ ($Q \equiv 0$ on the shadow kernel, verified at every filtration level, every prime). These are the **twelve** of the Watermark's Lemma C, returning as the carrier space of the form;
2. has shadow image of codimension 6 per family, cut by the **trinity relations**: three mass equalities, two linear compatibilities, and **the polarization identity** $(k-l)^2 + (k+l)^2 = 2k^2 + 2l^2$ — Lemma C's lone quadratic survivor, now closing the dimension count: $\dim(\text{image at moment-level } K) = 12m - 18 - 3K(K+1)/2$, exact at three primes;
3. is **perfect on the assembled shadow space** (axis self-pairings are standard one-variable inner products; the cross channels are hyperbolic pairs), with $W^\perp \subseteq W$ — deficit equal to the relation count, exact at three primes.

These three facts give, for the moment filtration m^K (zero moments up to total degree $< K$):

The ladder. $\text{rank}(Q|m^K) = 2 \cdot \dim(\text{image}) - 12m = 12(m-3) - 3K(K+1)$ for $m \geq 7$ (clamped at 0). Measured rungs: 48/42/30/12/0 ($m = 7$) and 96/90/78/60/36 ($m = 11$), every rung exact.

H_{top} is precisely the level $K = 2$, whence $\rho_2 = 12m - 54$ for $m \geq 7$. The drop per rung is $6K = 2 \times 3K$; the factor 2 is the rank of a Toeplitz block $u_i \cdot u_j \equiv (m-7)/2 - (i+j)$ discovered en route (instrument GOLLUM_PROBE_9) — the two halves of the proof are one mechanism, and the theorem takes its name from this: a double mechanism, read on a ladder.

The $m = 5$ boundary, stated once. At $m = 5$ a relation collision at the boundary frees $+2$ uniformly across the entire ladder (26/20/8/0/0 against the clamped law), giving $\rho_2 = \alpha = 8$: the form B_2 is *nondegenerate* on H_{top} . The same rank-saturation frontier appears in all three places the small prime deviates (the profile, the ladder, the radical) — it is one phenomenon, not three exceptions.

7. The arithmetic close

For $m \geq 7$: $b = t - 2\alpha + \rho_2 = 3(m-2)(m-4) - (3m^2 - 9m - 14) + (12m - 54) = 3m - 16$. For $m = 5$: $b = 9 - 16 + 8 = 1$. With $a = 3(m-3)^2 - 2b$: $a = 3m^2 - 24m + 59$ (resp. 10). ■

8. Corollaries

1. **The transcendental lattice.** $H^2(S_m, \mathbb{Z})$ is unimodular, so $T_m := NS^\perp$ has $T_m^*/T_m \cong V^*/V$ — the same group $(\mathbb{Z}/m)^{3m^2-24m+59} \times (\mathbb{Z}/m^2)^{3m-16}$ on rank $(m-1)(m-3)^2$. With the Watermark order this pins the genus data of both lattices to the level of the discriminant group.
2. **Shioda's Q7.4 for prime degree.** Question 7.4 of [7] asks for the determinant, and the Watermark Theorem answers it; this theorem adds the discriminant group, which the question did not ask for.

3. **Brauer groups.** The Artin–Tate recipe of the companion Brauer document now carries not only the order of the discriminant but its group — sharpening the $|Br|$ computations for the finite-field reductions.

9. Provenance and discipline of record

The operation ("Gollum", commissioned in the campaign's findings master §84) ran as eleven instruments (GOLLUM_PROBE_1–11, this repository), every one a full executable reproduced from file in a second, independent environment before ratification, under a standing dual-instance protocol (Constructor / Auditor) with the Architect as courier. Every law above was *declared before being measured* wherever a new cell was touched; the $m = 11$ profile $\{0:96, 1:158, 2:17\}$ was sealed in writing before the computation and met exactly (the same profile is row $(2, 11)$ of [6, Table 1]). The operation's graveyard is part of the record: a mixed-frame normalization bug (coordinates rescaled, form left unrescaled — exposed precisely by the 3×3 top blocks at $m \geq 7$), a Gram-route failure for independence whose autopsy yielded the Toeplitz block of §6, one $(1/m)\mathbb{Z}/\mathbb{Z}$ bookkeeping confusion, one integer-Smith overflow (replaced by pure mod- m^2 elimination), and two earlier heuristics killed by their own pre-registered falsifiers. Two consistency gates close the record: the formula-blind end-to-end computation of §3, and the independent Lemma-D route reproducing the full profiles at all three primes.

References

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Companion documents in this repository: THE_WATERMARK_THEOREM (the order) · THE_BRAUER_COROLLARY (the recipe) · THE_WATERMARK_LAW (the empirical campaign that began it).